

Visitor Management System using ServiceNow - Project Documentation

1. Introduction

Project Title: Visitor Management System using ServiceNow

2. Project Overview

Purpose: To automate visitor registration, badge generation, host notification, and entry/exit tracking using the ServiceNow platform.

Features: Visitor pre-registration via Service Catalog, automated badge/QR generation, host notifications, check-in/check-out tracking, dashboards, and visitor logs.

The Visitor Management System (VMS) is a ServiceNow-based automation solution designed to digitize and streamline the process of visitor registration, identity verification, badge issuance, and exit tracking within an organization. The system eliminates manual sign-in processes and reduces front-desk workload by providing a fully automated, self-service visitor experience.

The platform is built using the Service Catalog framework, where hosts can pre-register visitors by providing details such as visitor name, contact information, purpose of visit, expected arrival time, and host employee information. Upon submission, the system automatically notifies the host and the front desk, ensuring readiness before the visitor arrives.

Once the visitor checks in at the reception, the system generates a unique QR code-based visitor badge containing the visitor's name, visit purpose, host details, and validity duration. This digital badge acts as an access pass for controlled entry into the premises.

To enhance monitoring and operational visibility, the project includes real-time dashboards displaying visitor statistics, active visits, expected arrivals, and departure logs. Automated email notifications are sent to hosts upon visitor arrival, and to visitors upon successful check-out.

Overall, the solution demonstrates how ServiceNow can be used beyond IT service management to build real-world facility and security automation systems with workflow orchestration, scripting, and data visualization capabilities.

3. Architecture

Frontend: ServiceNow Service Portal / UI Pages

Backend: ServiceNow Business Rules, Script Includes, Client Scripts

Database: ServiceNow Tables (Custom Visitor Table / RITM), storing visitor and host data

The Visitor Management System is built on a layered ServiceNow architecture that separates user interaction, business logic, and data storage to ensure scalability, maintainability, and automation efficiency.

1. Presentation Layer (Frontend)

The frontend is implemented using ServiceNow Service Portal and UI Pages, which provide the user interface for hosts to pre-register visitors. Users interact with dynamic forms where they enter visitor details such as name, contact, visit purpose, expected arrival, and number of visitors. Client Scripts are used on the frontend to perform real-time validations and to enhance user experience, such as validating inputs and dynamically populating host information.

2. Application Logic Layer (Backend Processing)

The backend layer is responsible for all business logic and automation. It is implemented using:

Business Rules: Triggered automatically when records are inserted or updated. They handle badge generation, visitor ID generation, and host notification after visitor check-in.

Script Includes: Reusable server-side JavaScript modules used for complex logic such as badge generation and QR code creation logic.

Workflow / Flow Designer (if used): Orchestrates the visitor lifecycle from pre-registration to check-in, check-out, and notification delivery.

This layer ensures that all business rules are consistently enforced regardless of how the visitor entry is created.

3. Data Layer (Database)

All visitor data is stored in ServiceNow Tables, primarily:

A custom visitor management table (e.g., u_visitor_management) or the RITM (Requested Item) table depending on implementation

Associated tables for host details, visit logs, and badge records

These tables store structured data including visitor ID, visitor name, host employee, visit purpose, check-in time, check-out time, and generated QR badge reference

4. Integration & Output Layer

Once a visitor is successfully checked in, the system generates a QR code-based badge containing encoded visitor details, which is stored in the record and sent to the visitor and host via email notification. The Notification Engine of ServiceNow handles automated email delivery.

5. Reporting & Monitoring Layer

Dashboards and reports are built using ServiceNow Reporting tools to provide insights such as:

Total visitors checked in per day/week/month

Active visitors currently on premises

Most frequently visited departments or hosts

Average visit duration trends

4. Setup Instructions

Prerequisites: ServiceNow Developer Instance

Installation: Configure service catalog items, create visitor table, configure business rules and client scripts.

Step 1: Create Service Catalog Item

- Navigate to Service Catalog → Maintain Items
- Create a new catalog item named Visitor Pre-Registration
- Add required variables such as:
 - o Visitor Name (Text Field)
 - o Visitor Email / Phone (Text Field)
 - o Purpose of Visit (Dropdown)
 - o Expected Arrival Date & Time (Date/Time Picker)
 - o Host Employee Name (Reference to sys_user)
 - o Number of Visitors (Number Field)

Step 2: Create Backend Data Structure

- Navigate to System Definition → Tables
- Create a custom table (e.g., u_visitor_management) OR use RITM table (sc_req_item) if extending catalog flow
- Add required fields:
 - o Visitor ID (Auto-generated)
 - o Visitor Name
 - o Host Employee (Reference)

- o Visit Purpose
- o Check-In Time
- o Check-Out Time
- o Badge / QR Code (Image or URL field)
- o Visit Status (Active / Checked Out / Pre-Registered)

Step 3: Configure Client Scripts

- Navigate to Catalog Client Scripts
- Implement scripts for:
 - o Field validation (e.g., future date for expected arrival)
 - o Auto-populate host details on name selection
 - o Mandatory field enforcement before submission

Step 4: Configure Business Rules

- Navigate to System Definition → Business Rules
- Create rules to trigger after insert/update on visitor table
- Implement logic for:
 - o Automatic visitor ID generation
 - o Check-in timestamp capture
 - o Triggering QR badge generation process
 - o Host notification dispatch on visitor arrival

Step 5: Implement Script Includes

- Create reusable Script Includes for backend logic such as:
 - o Badge generation engine
 - o QR code generation function
 - o Utility methods for check-in/check-out processing

Step 6: Configure Notifications

- Navigate to System Notification → Email → Notifications
- Create notification triggered on:
 - o Pre-registration confirmation (to host)
 - o Visitor check-in (to host and security)
 - o Visitor check-out (to host)
- Attach visitor badge and QR code in check-in notification
- Ensure recipients are mapped to host employee and front desk

Step 7: Configure Reporting and Dashboard

- Navigate to Reports → Create New Report

- Build dashboards showing:
 - o Total visitors today and this week
 - o Active visitors currently on site
 - o Host-wise visitor frequency
- Add widgets to Service Portal for real-time front desk visibility

Step 8: Testing the System

- Submit a test visitor pre-registration via Service Portal
- Verify:
 - o Record creation in visitor table/RITM
 - o Host notification delivery on pre-registration
 - o QR badge generation on check-in
 - o Email notification with badge on check-in
 - o Check-out flow and dashboard data update

Final Note

Once all components are configured, the system is ready to handle end-to-end visitor registration and management through the ServiceNow platform.

5. Folder / Component Structure

1. Service Catalog Layer

- Contains the Visitor Pre-Registration Catalog Item
- Defines user-facing form variables such as:
 - o Visitor Name and Contact
 - o Host Employee
 - o Purpose of Visit
 - o Expected Arrival Date & Time
- Controls the user interaction flow in the Service Portal

2. Client-Side Logic

- Implemented using Client Scripts and UI Policies
- Responsible for:
 - o Real-time form validation (e.g., preventing past dates for arrival)
 - o Dynamic field behavior (mandatory fields, visibility conditions)
 - o Auto-population of host details on employee selection

3. Server-Side Business Logic

- Implemented using Business Rules and Script Includes

- Handles core backend operations such as:
 - o Visitor ID generation
 - o Check-in/check-out timestamp management
 - o QR badge generation logic trigger
- Ensures automation is executed consistently after record creation or update

4. Data Management Layer

- Uses ServiceNow Tables, either:
 - o Custom table (e.g., u_visitor_management)
 - o Or Service Catalog tables like sc_req_item
- Stores all transactional data including:
 - o Visitor details and contact information
 - o Host and department references
 - o Visit purpose and metadata
 - o Generated QR badge reference

5. Reporting & Analytics Layer

- Built using ServiceNow Reports and Dashboards
- Provides insights such as:
 - o Total visitors checked in
 - o Active visitors on premises
 - o Most frequently visited departments
 - o Daily and weekly visitor trends
- Supports operational monitoring and security decision-making

Key Structural Principle

The system follows a layered architecture model:

- UI Layer → Service Catalog + Portal
- Logic Layer → Client Scripts + Business Rules + Script Includes
- Data Layer → ServiceNow Tables
- Insight Layer → Reports & Dashboards

6. Running the Application

Access Service Portal URL → Submit pre-registration via catalog item →
Backend processes run automatically via Business Rules

Step 1: Access the Service Portal

Users open the ServiceNow Service Portal using the instance URL and navigate to the Visitor Pre-Registration catalog item available under the service catalog section.

Step 2: Fill Visitor Pre-Registration Form

The host enters the required visitor details, including:

- Visitor Name and Contact Information
- Purpose of Visit
- Expected Arrival Date and Time
- Number of Visitors

Client-side validations (implemented using Client Scripts and UI Policies) ensure:

- Mandatory fields are filled
- Arrival date is not in the past
- Input data format is correct before submission

Step 3: Submit Pre-Registration Request

Once the form is submitted, a service request is created in ServiceNow:

- A record is generated in the RITM or custom visitor table
- The request is assigned a unique Visitor ID automatically

Step 4: Backend Processing

After record creation, backend automation is triggered:

- Business Rules execute automatically on insert/update
- Visitor ID and metadata are finalized and stored in the database
- Script Includes handle reusable logic such as badge generation and QR code creation

Step 5: QR Badge Generation

The system generates a unique QR badge for the visitor, which contains encoded information such as:

- Visitor ID
- Visitor Name and Contact
- Host Employee Name and Department
- Visit Purpose and Date
- Badge Validity Duration

This QR badge is stored in the visitor record and linked to the user's check-in.

Step 6: Notification Delivery

Once processing is complete:

- An automated email notification is triggered
- The host receives visitor arrival confirmation with badge details and QR code

Step 7: Monitoring and Verification

Admins or users can track the visit through:

- ServiceNow record view (RITM or visitor table)
- Reports and dashboards showing visitor status and analytics

7. API / Automation Logic

The Visitor Management System primarily uses ServiceNow's internal automation framework. Most of the business logic is implemented using Business Rules and Script Includes, ensuring that visitor processing, badge generation, and workflow execution are handled within the platform.

However, for QR code generation, an external service is integrated to generate machine-readable visitor badge identifiers.

1. Internal Automation (ServiceNow Platform)

The following operations are handled natively within ServiceNow:

- Triggering Business Rules on record creation and update
- Visitor ID generation and record management
- Workflow execution for visitor processing lifecycle
- Notification triggering after successful check-in

These operations ensure that the system remains fully automated without manual intervention.

2. External QR Code Generation API

To generate QR codes for visitor badges, an external QR code generation service is integrated within a Business Rule or Script Include.

Working Flow:

- After check-in, the system sends encoded visitor data (Visitor ID, Name, Host, Department, Date, Validity) to the QR API
- The API returns a QR image URL or base64 encoded image
- This QR image is stored in the ServiceNow visitor record

- The QR code is included in the email notification sent to the host and visitor

3. Automation Trigger Point

- The QR badge generation process is triggered after update Business Rule on check-in
- This ensures the QR badge is generated only after a valid check-in is confirmed

4. Data Flow Summary

ServiceNow Visitor Record → Business Rule Trigger → Script Include Processing → QR API Call → QR Image Returned → Stored in Visitor Record → Notification Sent

8. Authentication

The Visitor Management System leverages ServiceNow's built-in Role-Based Access Control (RBAC) framework to ensure secure access to application data and functionalities.

1. Authentication Mechanism

User authentication is managed by the ServiceNow platform itself. Users must log in using valid ServiceNow credentials before accessing the Service Portal or backend modules. Session management, login validation, and user identity verification are handled by the platform's security layer.

2. Role-Based Access Control (RBAC)

Access to different parts of the system is restricted using predefined and custom roles.

Typical roles include:

- Host Role (itil or custom vms_host)
 - o Can access Service Portal to pre-register visitors
 - o Can view their own visitor bookings and check-in status
- Receptionist Role (custom vms_receptionist)
 - o Can manage visitor check-in and check-out at front desk
 - o Can view all expected visitors for the day
- Admin Role (admin)
 - o Full access to configuration, tables, scripts, and system settings
- Security Role (optional vms_security)
 - o Can view active visitors and badge validity

- o Can flag or block unauthorized visitor attempts

3. Data Security Controls

Security rules are applied at multiple levels:

- Table-level security (ACLs): Restrict access to visitor records based on user roles
- Field-level security: Sensitive fields (e.g., internal processing data) are hidden from end users
- Record-level security: Hosts can only view visitors they have registered
- Script protection: Server-side scripts ensure business logic cannot be bypassed from the frontend

4. Portal Security

- Service Portal access requires authentication
- Unauthorized users are redirected to login page
- Form submissions are validated server-side to prevent unauthorized data manipulation

5. QR Code Access Control

Generated QR codes are tied to authenticated visitor records, ensuring:

- Only valid check-ins generate valid QR badges
- QR data cannot be modified by end users after generation
- Expired badges (post check-out) are invalidated automatically

9. User Interface

9. User Interface

The Visitor Management System provides a user-friendly interface built using ServiceNow Service Portal, designed to simplify visitor registration, improve usability, and present real-time visitor information.

1. Visitor Pre-Registration Form (Service Portal)

The primary interface is a structured Service Catalog form where hosts input visitor details.

Key features include:

- Clean and responsive form layout optimized for end-user interaction
- Input fields for Visitor Name, Contact, Purpose of Visit, Expected Arrival, and Host Details

- Dynamic validations using Client Scripts and UI Policies
- Conditional field behavior (e.g., mandatory validation, error prompts)
- Real-time feedback before submission to prevent incorrect entries

This ensures that hosts can complete visitor pre-registration in a single streamlined flow.

2. Visitor Check-In / Check-Out Interface

After visitor arrival, the receptionist interface allows:

- Search and lookup for pre-registered visitors
- One-click check-in with automatic timestamp capture
- QR badge display and email option for the visitor
- Check-out confirmation with visit duration summary

3. Visitor Confirmation Page

After successful check-in, users are redirected to a confirmation page displaying:

- Unique Visitor ID
- Visitor and host details (name, department, contact)
- Visit purpose and duration
- Generated QR badge for access

The QR badge is prominently displayed and serves as the digital pass for controlled entry.

4. Dashboard Interface

An analytics dashboard is implemented using ServiceNow Reporting tools to provide operational insights.

It includes:

- Total visitors checked in over time
- Active visitors currently on premises
- Expected arrivals for the day
- Host-wise and department-wise visitor distribution
- Visual charts and graphs for quick interpretation

This helps administrators monitor system performance and visitor demand patterns.

5. User Experience Design

The interface is designed with the following principles:

- Simplicity: minimal steps required to register a visitor
- Consistency: uniform layout across portal pages
- Responsiveness: compatible with different screen sizes
- Clarity: clear error messages and confirmations

10. Testing

The Visitor Management System was validated using a structured manual testing approach to ensure correctness of visitor registration flow, badge generation logic, and check-in/check-out processing.

1. Test Scope

Testing was performed across the following core modules:

- Service Catalog Visitor Pre-Registration Form
- Visitor Check-In and Check-Out Flow
- Business Rules Execution
- QR Badge Generation
- Notification Delivery
- Dashboard Data Updates

2. Test Scenarios and Cases

Test Case 1: Visitor Pre-Registration Submission

- Input: Valid visitor name, host, purpose, and expected arrival
- Expected Output: Visitor record is created successfully in ServiceNow table/RITM
- Result: Pass

Test Case 2: Input Validation

- Input: Past date selected for expected arrival
- Expected Output: System should block submission and show validation error
- Result: Pass

Test Case 3: Visitor Check-In

- Input: Pre-registered visitor arriving at front desk
- Expected Output: Check-in timestamp recorded and Business Rule triggered

- Result: Pass

Test Case 4: Visitor ID Generation

- Input: New visitor check-in submission
- Expected Output: Unique visitor ID generated automatically
- Result: Pass

Test Case 5: QR Badge Generation

- Input: Successfully checked-in visitor record
- Expected Output: QR badge generated using Script Include and external API integration
- Result: Pass

Test Case 6: Notification Delivery

- Input: Successful visitor check-in
- Expected Output: Email notification sent to host with visitor details and QR badge
- Result: Pass

11. Known Issues

Despite successful implementation of the Visitor Management System, certain limitations exist due to platform constraints and external dependencies.

1. Dependency on External QR Code API

QR badge generation relies on an external service. If the API is unavailable or slow, badge generation and visitor confirmation may be delayed or fail. This introduces a dependency outside ServiceNow's control.

2. Limited Offline Validation

The system requires active connectivity to ServiceNow for QR verification and visitor validation. Offline scanning or verification at entry points is not supported.

3. No Biometric or ID Verification

The system does not include identity verification beyond the registration form. There is no integration with ID scanning or biometric systems to confirm visitor identity at check-in.

4. No Real-Time Access Control Integration

The system does not currently integrate with physical access control systems (e.g., turnstiles, door locks). Badge validity is managed digitally only and not enforced through hardware.

5. Basic Fraud Protection

QR codes contain encoded visitor data but do not include advanced encryption or blockchain-level validation, making them theoretically vulnerable to duplication if not controlled properly.

6. Limited Scalability Testing

The system has not been tested under high-load conditions (e.g., large-scale concurrent visitor registrations), so performance under heavy traffic is not validated.

12. Future Enhancements

The Visitor Management System can be extended further to improve usability, scalability, and real-world deployment readiness.

1. Mobile QR Scanner Integration

A dedicated mobile scanning module can be developed for entry points. This would allow real-time scanning and validation of visitor QR badges, enabling faster verification and reducing manual ticket checks.

2. Biometric Identity Verification

Integration with fingerprint or facial recognition systems would add an additional layer of identity assurance, ensuring that the registered visitor is the same person arriving at the premises.

3. Physical Access Control Integration

Integration with turnstiles, door sensors, and badge readers would enable automatic entry and exit based on QR badge scan, removing the need for manual front-desk intervention.

4. Visitor Self-Check-In Kiosk

A kiosk-based self-service interface at reception would allow pre-registered visitors to check themselves in, retrieve their QR badge, and notify their host automatically, reducing front desk workload significantly.

5. Real-Time SMS Notifications

Adding SMS gateway integration alongside email would enable instant text alerts to hosts and visitors for check-in, check-out, and unexpected visitor arrivals.